

```
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1) {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 4) :orclause <> :rinfo :serial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12)}} :outer_is_left true :hashjoinoperator 96 :left_bucketsize 0.0058823529411764705 :right_bucketsize 1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hasheqoperator 96 :right_hasheqoperator 96))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1) {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 4) :orclause <> :rinfo :serial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12)}} :outer_is_left true :hashjoinoperator 96 :left_bucketsize 0.0058823529411764705 :right_bucketsize 1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 10 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 10 :varattnosyn 1 :location -1) {VAR :varno 7 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
OPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((AGGREF :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {VAR :varno 5 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtranso 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((AGGREF :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {VAR :varno 5 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtranso 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 5 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 5 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 After pull up At starting of Path Generation:
DETAIL:
:QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname part
        :colnames ("p_partkey" "p_name" "p_mfgr" "p_brand" "p_type" "p_size" "p_container" "p_retailprice" "p_comment")
      }
    :rtekind 0
    :reld 17169
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier_secure
        :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
      }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :reld 17274
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 2
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp_secure
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p_s_comment")
      }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :reld 17278
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 3
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname nation
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
    :rtekind 0
    :reld 17158
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 4
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname region
        :colnames ("r_regionkey" "r_name" "r_comment")
      }
    :rtekind 0
    :reld 17164
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 5
    :tablesample <>
    :lateral false
```

```

:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17191
:inh false
:relkind r
:rellockmode 1
:perminfoindex 9
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17274
:inh false
:relkind v
:rellockmode 1
:perminfoindex 10
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 11
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS

```

```

        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
}
:rtkind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 12
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtkind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 13
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rtperminfos (
{RTEPERMISSIONINFO
:relid 17169
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17274
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17278
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17274
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
}
{RTEPERMISSIONINFO
:relid 17164

```

```

:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 6
}
)
:quals (
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 7
}
{RANGETBLREF
:rtindex 8
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
{VAR
:varno 6
:varattno 4
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 4
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
{VAR
:varno 8
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 1
:location -1
}
)
}
)

```

supplier SEMI JOIN
nation

First view tables

s.s_acctbal > 0

n.n_nationkey =
s.s_nationkey

n.n_regionkey =
r.r_regionkey

```

:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
:varno 8
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 2
:location -1
}
  {CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
}
:alias <>
:rtindex 0
}
JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 9
}
)
:quals (
  {OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 9
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 3
:location -1
}
  {CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:rarg
JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 11
}
)
:quals (
  {OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 11
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 6
:location -1
}
  {CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
}
}
:rarg
{FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 12
}
  {RANGETBLREF
:rtindex 13
}
)
:quals <>
}
:usingClause <>

```

r.r_name = 'EUROPE'

partsupp ps
JOIN supplier s

ps.ps_availqty > 100

supplier s (right side)
same join as above

s.s_acctbal > 0

```

:join_using_alias <>
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 12
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 12
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 11
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 4
        :location -1
      }
    )
    :location -1
  }
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 12
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 12
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 13
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 13
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
  {OPEXPR
    :opno 1054
    :opfuncid 1048
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
      {VAR
        :varno 13
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 13
        :varattnosyn 2
        :location -1
      }
      {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
      }
    )
    :location -1
  }
)
:alias <>
:rtindex 0
}
:usingClause <>
:join_using_alias <>
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 11
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 9
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 9

```

n.n_nationkey =
s.s_nationkey

n.n_regionkey =
r.r_regionkey

r.r_name = 'EUROPE'

s.s_suppkey =
ps.ps_suppkey

```

        :varattnosyn 2
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 11
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 6
        :location -1
    }
    {CONST
        :consttype 1700
        :consttypmod -1
        :constcollid 0
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 6 [ 24 0 0 0 0 -128 ]
    }
}
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
:equals (
    {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
            {VAR
                :varno 1
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 1
                :location 297
            }
            {VAR
                :varno 9
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 9
                :varattnosyn 1
                :location -1
            }
        )
        :location 307
    }
    {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
            {VAR
                :varno 6
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 6
                :varattnosyn 1
                :location -1
            }
            {VAR
                :varno 9
                :varattno 2
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 9
                :varattnosyn 2
                :location -1
            }
        )
        :location 342
    }
    {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
            {VAR
                :varno 1
                :varattno 6
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 6
                :location 367
            }
        )
    }
    {CONST

```

s.s_acctbal > 0

WHERE

p_partkey =
 ps_partkey
 AND s_suppkey =
 ps_suppkey
 AND p_size = 15
 AND p_type LIKE
 '%BRASS'
 AND s_nationkey =
 n_nationkey
 AND n_regionkey =
 region.r_regionkey
 AND region.r_name
 = 'EUROPE'

Main query Condition

```

:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 376
:constvalue 4 [ 15 0 0 0 0 0 0 ]
}
)
:location 374
}
}
COEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {RELABELTYPE
    :arg
      {VAR
        :varno 1
        :varattno 5
        :vartype 1043
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 5
        :location 391
      }
      :resulttype 25
      :resulttypmod -1
      :resultcollid 100
      :relabelformat 2
      :location -1
    }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 403
    :constvalue 10 [ 40 0 0 0 0 37 66 82 65 83 83 ]
  }
)
:location 398
}
}
COEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 6
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 4
    :location -1
  }
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 438
  }
)
:location 436
}
}
COEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 4
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 3
    :location 462
  }
  {VAR
    :varno 5
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 1
    :location 476
  }
)
:location 474
}
}
COEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 5
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 2
    :location 507
  }
)

```

```

    }
    {CONST
    :consttype 1042
    :consttypemod -1
    :constcollid 100
    :consttlen -1
    :constbyval false
    :constisnull false
    :location 523
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
  )
  :location 521
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
  :varno 9
  :varattno 4
  :vartype 1700
  :vartypmod 983046
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 9
  :varattnosyn 4
  :location -1
  }
  {SUBPLAN
  :sublinkType 4
  :testexpr <>
  :paramids <>
  :plan_id 1
  :plan_name SubPlan\ 1
  :firstCollType 1700
  :firstCollTypemod -1
  :firstCollCollation 0
  :useHashTable false
  :unknownEqFalse false
  :parallel_safe false
  :setParam <>
  :parParam (i 0)
  :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 840
    }
  )
  :startup_cost 0
  :per_call_cost 15254.967750000003
  }
)
:location 558
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
  :expr
  {VAR
  :varno 6
  :varattno 6
  :vartype 1700
  :vartypmod 983046
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 6
  :varattnosyn 6
  :location -1
  }
  }
  :resno 1
  :resname s_acctbal
  :ressortgroupref 1
  :resorigtbl 17274
  :resorigcol 6
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 6
  :varattno 2
  :vartype 1042
  :vartypmod 29
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 6
  :varattnosyn 2
  :location -1
  }
  }
  :resno 2
  :resname s_name
  :ressortgroupref 3
  :resorigtbl 17274
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 4
  :varattno 2
  :vartype 1042
  :vartypmod 29
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 4
  :varattnosyn 2
  :location 86
  }
  }
  :resno 3
  :resname n_name
  :ressortgroupref 2
  :resorigtbl 17158
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 1
  :vartype 23

```

```

(
    select
    min(ps_supplycost)
    from
    partsupp_secure,
    supplier_secure,
    nation,
    region
    where
    p_partkey = ps_partkey      same as rls
    and
    s_suppkey =
    ps_suppkey
    and
    s_nationkey =
    n_nationkey
    and
    n_regionkey =
    region.r_regionkey
    and
    region.r_name =
    'EUROPE'
)

```

```

:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 17169
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 3
:varatype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location -1
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17274
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 5
:varatype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 5
:location -1
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17274
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 7
:varatype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 7
:location -1
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17274
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 3
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}

```

```

}
:limitOffset <=
:limitCount
{CONST
  :consttype 20
  :consttypemod -1
  :constcollid 0
  :constlen 8
  :constbval true
  :constnull false
  :location -1
  :constvalue 8 [ 100 0 0 0 0 0 0 0 ]
}
:limitOption 0
:rowMarks <=
:setOperations <=
:constraintDeps <=
:withCheckOptions <=
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1289 :ofuncid 850 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((RELABELTYPE :arg (VAR :varno 1 :varattno 5 :vartype 1043 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) :resulttype 25 :resulttypemod -1 :resultcollid 100 :relabeltype 2 :location -1) (CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen 1 :constbval true :constnull false :location -1) :constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <= :rinfo_serial 15 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 6 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1) :constvalue 4 [ 15 0 0 0 0 0 0 0 ]) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause (OPEXPR :opno 96 :ofuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1) :constvalue 4 [ 15 0 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <= :rinfo_serial 14 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 6 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1) :constvalue 4 [ 15 0 0 0 0 0 0 0 ]) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause (OPEXPR :opno 1289 :ofuncid 850 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((RELABELTYPE :arg (VAR :varno 1 :varattno 5 :vartype 1043 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) :resulttype 25 :resulttypemod -1 :resultcollid 100 :relabeltype 2 :location -1) (CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen 1 :constbval true :constnull false :location -1) :constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <= :rinfo_serial 15 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 6 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1) :constvalue 4 [ 15 0 0 0 0 0 0 0 ]) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (VAR :varno 1 :varattno 6 :vartype 1043 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) :is_parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 4 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) (VAR :varno 4 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) (VAR :varno 4 :varattno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 4 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) (VAR :varno 4 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) (VAR :varno 4 :varattno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1054 :ofuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 5 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 2 :location -1) (CONST :consttype 1042 :consttypemod -1 :constcollid 100 :constlen -1 :constbval false :constnull false :location -1) :constvalue 10 [ 40 0 0 69 85 82 79 80 69 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 5) :required_relds (b 5) :incompatible_relds (b) :outer_relds (b) :left_relds (b 5) :right_relds (b) :orclause <= :rinfo_serial 22 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 2 :location -1) :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 1042 :consttypemod -1 :constcollid 100 :constlen -1 :constbval false :constnull false :location -1) :constvalue 10 [ 40 0 0 69 85 82 79 80 69 ])) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (CONST :consttype 1042 :consttypemod -1 :constcollid 100 :constlen -1 :constbval false :constnull false :location -1) :constvalue 10 [ 40 0 0 69 85
```

[illegible]

[illegible]

rattno 7 :vartype 1043 :vartypemod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 6 :varattno 6 :vartype 1700 :vartypemod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 6 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 6 :varattno 6 :vartype 1700 :vartypemod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 6 :varattno 5 :vartype 1042 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 5 :location -1} {VAR :varno 6 :varattno 7 :vartype 1043 :vartypemod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 7 :location -1}) | parallel_safe = true | reason = PROPAGATE_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 6 :varattno 6 :vartype 1700 :vartypemod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 6 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 6 :varattno 6 :vartype 1700 :vartypemod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 6 :varattno 5 :vartype 1042 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 5 :location -1} {VAR :varno 6 :varattno 7 :vartype 1043 :vartypemod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 6 :varattno 6 :vartype 1700 :vartypemod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 6 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 6 :varattno 6 :vartype 1700 :vartypemod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 6 :varattno 5 :vartype 1042 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 5 :location -1} {VAR :varno 6 :varattno 7 :vartype 1043 :vartypemod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 7 :location -1}) | parallel_safe = true | reason = PROPAGATE_SAFE
QUERY PLAN

Limit (cost=23999.82..23999.83 rows=1 width=271) (actual time=21669.293..21669.383 rows=100.00 loops=1)
Output: s.s.acctbal, s.s.name, nation.n.name, part.p.partkey, part.p.mfgr, s.s.address, s.s.phone, s.s.comment
Buffers: shared hit=4950615
-> Sort (cost=23999.82..23999.83 rows=1 width=271) (actual time=21669.293..21669.376 rows=100.00 loops=1)
Output: s.s.acctbal, s.s.name, nation.n.name, part.p.partkey, part.p.mfgr, s.s.address, s.s.phone, s.s.comment
Sort Key: s.s.acctbal DESC, nation.n.name, s.s.name, part.p.partkey
Sort Method: quicksort Memory: 77kB
Buffers: shared hit=4950615
-> Hash Join (cost=5036.89..23999.81 rows=1 width=271) (actual time=14166.148..21668.973 rows=189.00 loops=1)
Output: s.s.acctbal, s.s.name, nation.n.name, part.p.partkey, part.p.mfgr, s.s.address, s.s.phone, s.s.comment
Hash Cond: ((ps.ps.partkey = part.p.partkey) AND (ps.ps.supplycost = (SubPlan 1)))
Buffers: shared hit=4950606
-> Nested Loop (cost=1449.90..12785.32 rows=1 width=251) (actual time=3.816..26.291 rows=70201.00 loops=1)
Output: s.s.acctbal, s.s.name, s.s.address, s.s.phone, s.s.comment, ps.ps.partkey, ps.ps.supplycost, nation.n.name
Join Filter: (nation.n.regionkey = region.r.regionkey)
Buffers: shared hit=9480
-> Seq Scan on public.region (cost=0.00..12.12 rows=1 width=4) (actual time=0.012..0.023 rows=1.00 loops=1)
Output: region.r.regionkey, region.r.name, region.r.comment
Filter: (region.r.name = 'EUROPE')::bpchar
Rows Removed by Filter: 4
Buffers: shared hit=1
-> Gather (cost=1449.90..12773.04 rows=12 width=255) (actual time=3.801..14.723 rows=70201.00 loops=1)
Output: s.s.acctbal, s.s.name, s.s.address, s.s.phone, s.s.comment, ps.ps.partkey, ps.ps.supplycost, nation.n.name, nation.n.regionkey
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=9479
-> Hash Join (cost=449.90..11771.84 rows=5 width=255) (actual time=3.430..39.140 rows=23400.33 loops=3)
Output: s.s.acctbal, s.s.name, s.s.address, s.s.phone, s.s.comment, ps.ps.partkey, ps.ps.supplycost, nation.n.name, nation.n.regionkey
Hash Cond: (s_1.s.supkey = s.s.supkey)
Buffers: shared hit=9479
Worker 0: actual time=3.414..52.152 rows=32009.00 loops=1
Buffers: shared hit=4228
Worker 1: actual time=3.377..56.505 rows=34595.00 loops=1
Buffers: shared hit=4570
-> Hash Semi Join (cost=224.10..11542.66 rows=891 width=18) (actual time=1.710..30.898 rows=23400.33 loops=3)
Output: ps.ps.supkey, ps.ps.partkey, ps.ps.supplycost, s_1.s.supkey
Hash Cond: (ps.ps.supkey = s_1.s.supkey)
Buffers: shared hit=9134
Worker 0: actual time=1.724..41.631 rows=32009.00 loops=1
Buffers: shared hit=4113
Worker 1: actual time=1.720..45.243 rows=34595.00 loops=1
Buffers: shared hit=4455
-> Parallel Seq Scan on public.partsupp ps (cost=0.00..10875.33 rows=165071 width=14) (actual time=0.000..16.956 rows=131975.33 loops=3)
Output: ps.ps.partkey, ps.ps.supkey, ps.ps.availqty, ps.ps.supplycost, ps.ps.comment
Filter: (ps.ps.availqty > 100)
Rows Removed by Filter: 1358
Buffers: shared hit=8792
Worker 0: actual time=0.005..23.297 rows=179864.00 loops=1
Buffers: shared hit=3999
Worker 1: actual time=0.006..25.449 rows=195632.00 loops=1
Buffers: shared hit=4341
-> Hash (cost=223.76..223.76 rows=27 width=4) (actual time=1.693..1.697 rows=887.00 loops=3)
Output: s_1.s.supkey
Buckets: 1024 Batches: 1 Memory Usage: 40KB
Buffers: shared hit=342
Worker 0: actual time=1.709..1.714 rows=887.00 loops=1
Buffers: shared hit=114
Worker 1: actual time=1.704..1.707 rows=887.00 loops=1
Buffers: shared hit=114
-> Hash Semi Join (cost=24.50..223.76 rows=27 width=4) (actual time=0.124..1.594 rows=887.00 loops=3)
Output: s_1.s.supkey
Hash Cond: (s_1.s.nationkey = n_1.n.nationkey)
Buffers: shared hit=342
Worker 0: actual time=0.169..1.612 rows=887.00 loops=1
Buffers: shared hit=114
Worker 1: actual time=0.163..1.608 rows=887.00 loops=1
Buffers: shared hit=114
-> Seq Scan on public.supplier s_1 (cost=0.00..187.00 rows=4558 width=8) (actual time=0.040..1.194 rows=4566.00 loops=3)
Output: s_1.s.supkey, s_1.s.name, s_1.s.address, s_1.s.nationkey, s_1.s.phone, s_1.s.acctbal, s_1.s.comment
Filter: ((s_1.s.acctbal > '0')::numeric) AND (s_1.s.acctbal > '0')::numeric)
Rows Removed by Filter: 434
Buffers: shared hit=336
Worker 0: actual time=0.061..1.194 rows=4566.00 loops=1
Buffers: shared hit=112
Worker 1: actual time=0.056..1.188 rows=4566.00 loops=1
Buffers: shared hit=112
-> Hash (cost=24.48..24.48 rows=1 width=4) (actual time=0.077..0.079 rows=5.00 loops=3)
Output: n_1.n.nationkey
Buckets: 1024 Batches: 1 Memory Usage: 9KB
Buffers: shared hit=6
Worker 0: actual time=0.100..0.102 rows=5.00 loops=1
Buffers: shared hit=2
Worker 1: actual time=0.100..0.102 rows=5.00 loops=1
Buffers: shared hit=2
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.068..0.074 rows=5.00 loops=3)
Output: n_1.n.nationkey
Hash Cond: (n_1.n.regionkey = r_1.r.regionkey)
Buffers: shared hit=6
Worker 0: actual time=0.092..0.097 rows=5.00 loops=1
Buffers: shared hit=2
Worker 1: actual time=0.091..0.097 rows=5.00 loops=1
Buffers: shared hit=2
-> Seq Scan on public.nation n_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.027..0.029 rows=25.00 loops=3)
Output: n_1.n.nationkey, n_1.n.name, n_1.n.regionkey, n_1.n.comment
Buffers: shared hit=3
Worker 0: actual time=0.038..0.039 rows=25.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.038..0.040 rows=25.00 loops=1
Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.033..0.034 rows=1.00 loops=3)
Output: r_1.r.regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9KB
Buffers: shared hit=3
Worker 0: actual time=0.046..0.047 rows=1.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.045..0.047 rows=1.00 loops=1
Buffers: shared hit=1
-> Seq Scan on public.region r_1 (cost=0.00..12.12 rows=1 width=4) (actual time=0.026..0.027 rows=1.00 loops=3)
Output: r_1.r.regionkey
Filter: (r_1.r.name = 'EUROPE')::bpchar
Rows Removed by Filter: 4
Buffers: shared hit=3
Worker 0: actual time=0.037..0.039 rows=1.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.037..0.040 rows=1.00 loops=1
Buffers: shared hit=1
-> Hash (cost=225.46..225.46 rows=27 width=249) (actual time=1.710..1.713 rows=887.00 loops=3)
Output: s.s.acctbal, s.s.name, s.s.address, s.s.phone, s.s.comment, s.s.supkey, nation.n.name, nation.n.regionkey
Buckets: 1024 Batches: 1 Memory Usage: 189KB
Buffers: shared hit=345
Worker 0: actual time=1.683..1.687 rows=887.00 loops=1
Buffers: shared hit=115
Worker 1: actual time=1.650..1.654 rows=887.00 loops=1
Buffers: shared hit=115